

Coupling-gated cortical coherence: a conditional framework for unity and self-representation

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6 September 2026

Abstract

How could coordinated cortical activity support one internally represented situation? We distinguish coherence, agreement between local contents, and accurate self-reconstruction, and develop conditional connections among them. Declared actuator modes give a necessary installed-energy condition for a coherent stationary branch. Phase encoders shared on overlaps bound content disagreement; compatible spatial measures form one global state. A contracting reconstruction that fixes this state makes it the unique fixed point, while accuracy across a state family requires distinguishable encodings. Lean 4 checks the results and their composition under explicit connecting hypotheses. Numerical controls delimit their observational use: finite ramps delay coherence, distinct mechanisms share stationary phase distributions, and content agreement requires independently validated decoders. We propose the global state and its internal reconstruction as correlates of unity and a minimal self. An extracellular-field hypothesis links awakening to increasing coupling. Its test requires calibrated geometry and phase dynamics, together with measurements of content agreement, self-encoding and experience.

1 Introduction

How could activity distributed across cortex support one experience? A field account proposes that endogenous extracellular electromagnetic fields partly coordinate spatially separated neural processes (McFadden, 2002, 2020; Pockett, 2012). Ephaptic effects provide a possible interaction mechanism (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015). Successful coordination would still leave a representational question: what makes the coordinated activity a description of one situation, with a perspective on that situation?

We distinguish three requirements. *Coherence* concerns collective phase order. *Compatibility* concerns whether local descriptions agree where their domains overlap. *Self-representation* concerns whether the resulting state carries an accurate internal reconstruction of itself. We propose the latter two relations as formal correlates of conscious unity and a minimal self; their interpretation requires evidence beyond the phase model.

Unity has been approached differently elsewhere: integrated information theory treats integration and exclusion as constitutive (Tononi et al., 2016), global-workspace accounts tie availability to broadcast (Baars, 1997; Dehaene et al., 2014), and predictive-processing accounts derive both from a hierarchical generative model (Friston, 2010). Here the three requirements are kept apart rather than derived from one another, so that each carries its own assumptions and admits its own measurement; Section 8.4 returns to the

23 comparison.

24 Consider reaching for a cup. A visual description of its location and a
25 description of the hand's relation to it can agree about where the cup is while
26 contributing different contents. Synchronizing the two processes would leave
27 disagreement about that shared quantity unresolved. An internal representa-
28 tion must additionally locate the perceiver within the represented situation.
29 This example motivates the requirements; the mathematical constructions
30 use spatial measures and fields in real normed vector spaces, whose identifi-
31 cation with perceptual content is a further modelling task.

32 The mathematical contribution is to specify connections between these
33 requirements and the assumptions each connection needs. A declared ac-
34 tuator's stored energy bounds its mean coupling. Phase encoders shared on
35 overlaps convert phase order into a bound on content disagreement. Compat-
36 ible spatial measures have a unique global extension; approximate agreement
37 controls a specified global selection. A contracting reconstruction makes that
38 extension its unique fixed point only if it accurately reconstructs the exten-
39 sion. Accuracy across a family of states imposes a further condition on the
40 information retained by the encoding.

41 The empirical contribution is to identify what observations can decide.
42 Stationary phase distributions can be shared by stronger coupling, weaker
43 diffusion and a matched common drive. Finite ramps change the observable
44 onset. A winding field is stationary at the same coupling as a uniform one and
45 has the opposite resultant, so that observable does not determine the state.

46 Coherent adaptation can lose alignment with a designated environmental
47 template, and a compatibility estimator needs diagnostics that distinguish
48 content from decoder artifacts. These findings determine the measurements
49 needed to test the proposed connections.

50 Sections 2–4 develop the three requirements together (Fig. 1). Section 5
51 composes them, stating the thermodynamic constraints and all eight con-
52 necting hypotheses per branch; its supporting models of feedback, learning
53 and resource supply have distinct state spaces and assumptions. Section 6
54 then examines what the observables can establish, before Section 7 devel-
55 ops the cortical hypothesis and an awakening experiment. The hypothesis is
56 that extracellular contraction increases field-mediated coupling; its direction,
57 rate and relation to recovery require independent calibration. Supplemen-
58 tal Material organizes the constructions and controls behind the claims they
59 support.

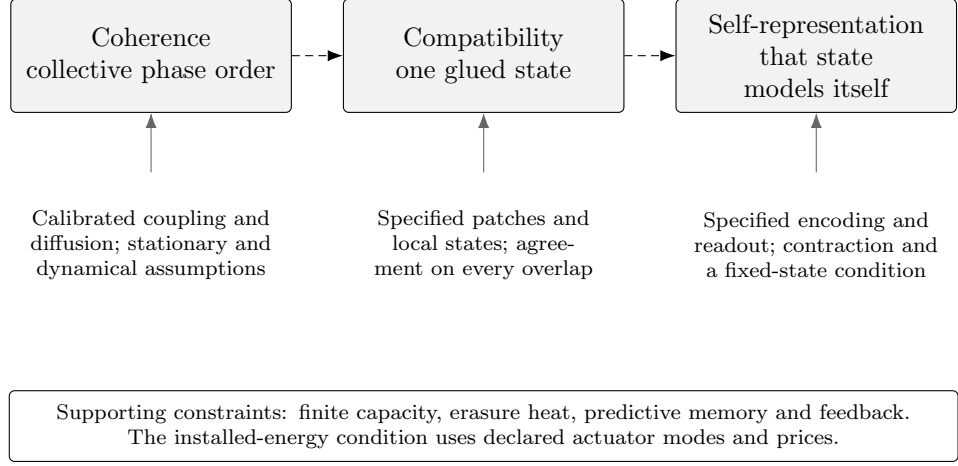


Figure 1: Three requirements of the proposed account. Each requires the inputs shown below it. Dashed arrows mark proposed connections: phase coherence alone supplies neither compatible local descriptions nor an accurate self-model. Thermodynamics constrains a candidate implementation through separate assumptions. This is the explanatory structure of the article; Table 1 records the eight implications consumed by each branch of the formal composition.

60 2 Coherence: physical coupling and its re- 61 source condition

62 2.1 Dynamics and the stationary threshold

63 The finite stochastic model is

$$d\theta_i = \left[\omega_i + \sum_j K_{ij}(t) \sin(\theta_j - \theta_i) \right] dt + \sqrt{2D} dW_i, \quad r_N = \left| \frac{1}{N} \sum_i e^{i\theta_i} \right|. \quad (1)$$

Here $D > 0$ is phase diffusion, W_i are independent Wiener processes, and K_{ij} has inverse-time units, with angles dimensionless. Uniform mean-field coupling is $K_{ij} = K/N$; frequency heterogeneity and diffusion are distinct parameters. The stationary calculation below uses identical frequencies and this scalar coupling. Applying it to a spatial kernel requires an aggregation rule and a check of the resulting approximation.

For identical frequencies in their rotating frame, the positive normalized stationary densities of the one-body Fokker–Planck equation are exactly

$$\rho(\theta) = \frac{e^{a \cos(\theta - \psi)}}{2\pi I_0(a)}, \quad a = \frac{Kr}{D}, \quad r = R(a) := \frac{I_1(a)}{I_0(a)}, \quad (2)$$

where I_n are modified Bessel functions and ψ is the collective phase. The formal derivation uses the absence of stationary current for a periodic gradient drift to obtain this density, then verifies normalization and self-consistency. At $K \leq K_c = 2D$, only $r = 0$ is self-consistent. Above threshold, exactly one positive solution coexists with zero and increases strictly with K . Its onset is

$$\frac{r^2}{K - K_c} \longrightarrow \frac{1}{D}, \quad r \sim \sqrt{\frac{K - K_c}{D}}. \quad (3)$$

Figure 2 shows this stationary branch.

Linearization about the uniform density gives rate $K/2 - D$ for the first harmonic and $-Dn^2$ for harmonics $n \geq 2$. A growing mode therefore exists exactly above threshold. About the coherent density, the quadratic form is strictly dissipative transverse to collective rotation, whose rate is zero.

83 These linear estimates concern the limiting equation. Relating them to finite
84 stochastic trajectories requires a dynamical mean-field limit and branch selec-
85 tion; exchange symmetry alone does not establish the needed independence
86 (Sznitman, 1991).

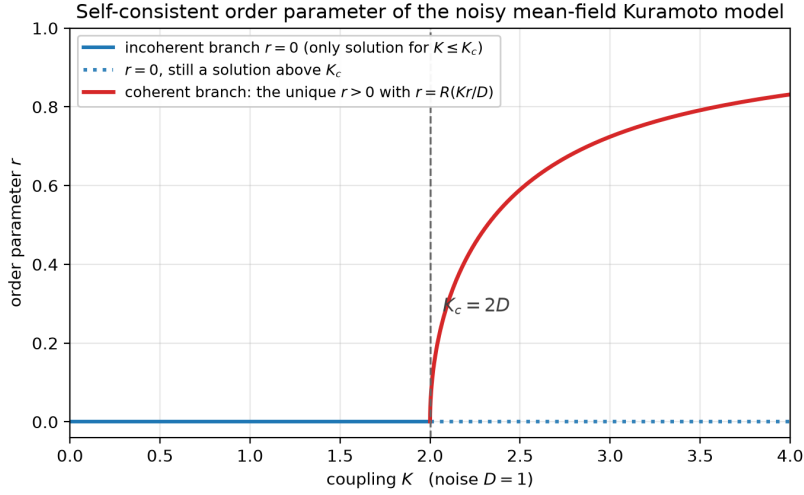


Figure 2: Stationary self-consistency for identical frequencies and phase diffusion $D = 1$. The positive coherent branch appears continuously above $K_c = 2D$ and increases with coupling. Zero remains a solution above threshold. The curve is computed from the integral expressions used in the formalization.

87 2.2 Installed energy bounds mean coupling

88 A spatial model replaces the finite sum by integration against a kernel. One
89 declared implementation carries response modes with continuous profiles ϕ_i ,

90 installation prices u_i and nonnegative occupancies $c_i(z)$ in configuration z :

$$K_z(x, y) = \sum_i c_i(z) \phi_i(x) \phi_i(y), \quad U(z) = \sum_i u_i c_i(z). \quad (4)$$

91 The same configuration sets the reciprocal kernel and stored energy. An
 92 executed transition changes the kernel through its changed occupancies and
 93 requires work $U(z') - U(z) + Q$, with Q positive into the reservoir. The
 94 installed kernel is averaged over the resulting configuration law $p(z)$. Cell-
 95 pair reconstruction on refining spatial partitions converges to its product-
 96 measure integral under the specified compactness and sampling assumptions;
 97 the refinement index is independent of the number of control updates.

98 Normalize the substrate measure μ to one. Write $m_i = \int \phi_i d\mu$ and
 99 declare $\kappa > 0$ with $m_i^2 \leq \kappa u_i$ for every mode. Factorizing each mode's
 100 integral gives

$$\begin{aligned} U_{\text{inst}} &= \sum_z p(z) U(z), \\ K_{\text{mf}} &= \iint K(x, y) d\mu(x) d\mu(y) = \sum_z p(z) \sum_i c_i(z) m_i^2 \leq \kappa U_{\text{inst}}. \end{aligned} \quad (5)$$

101 If this mean coupling is the K of the scalar model, $\kappa U_{\text{inst}} \leq 2D$ excludes a
 102 coherent stationary branch and any growing Fourier mode about the uniform
 103 density. Thus coherence on that branch requires $U_{\text{inst}} > 2D/\kappa$. The condition
 104 is necessary: energy above it need not produce sufficient coupling because
 105 the bound can be loose.

106 The quantity constrained is installed energy. Dissipated heat does not
 107 determine it, and κ is supplied by the mode profiles, prices and substrate
 108 normalization. Applying the scalar threshold to heterogeneous or signed spa-
 109 tial dynamics requires a justified reduction. Identifying the installed kernel
 110 with a cortical field is a separate physical commitment.

111 **3 Compatibility: from coordinated activity** 112 **to a global description**

113 **3.1 Why common timing is insufficient**

114 For overlapping patches U and V , compatibility means that restricting either
 115 local description to $U \cap V$ gives the same object. Descriptions can agree on
 116 shared quantities while contributing different information elsewhere. Com-
 117 patible sections of the specified sheaf have a unique global extension. On
 118 a finite cover, local finite spatial measures agreeing as measures on every
 119 overlap also have a unique finite-measure extension whose sheaf image is the
 120 glued section. No mass-one normalization is imposed.

121 In the cup example, visual and reaching descriptions must express a
 122 shared quantity, such as cup location, in common coordinates. Restriction
 123 retains that quantity while dropping details particular to each description.
 124 Agreement permits surface appearance and hand configuration to contribute
 125 to one situation. The mathematical counterexample assigns the same phase

126 to every patch while two local mass profiles disagree on a shared site. Those
 127 profiles have no common extension. Because they are chosen independently
 128 of phase, synchrony places no constraint on their disagreement.

129 This distinction limits the biological interpretation. Receptive territories
 130 and content domains must be related to spatial patches, and the descriptions
 131 must support the required extension. Matching overlap marginals of de-
 132 coded probability laws need not determine a unique global joint distribution
 133 (Abramsky & Brandenburger, 2011).

134 **3.2 When coherence does constrain content**

135 A declared relation between phase and content gives a positive result. Let
 136 contents take values in a real normed vector space E . Patch i reads an
 137 assigned oscillator through $e_{i,x}$ at site x , evaluated on the circle point
 138 $(\cos \theta, \sin \theta)$. Assume a uniform Lipschitz constant L in chord distance and
 139 require $e_{i,x} = e_{j,x}$ wherever patches i and j overlap. Encoders may vary across
 140 sites; their equality on overlaps is a supplied relation between descriptions of
 141 the shared quantity. The order-parameter identity $N^2 r^2 = \sum_{ij} \cos(\theta_i - \theta_j)$
 142 implies $1 - \cos(\theta_i - \theta_j) \leq N^2(1 - r^2)$ for each pair. Consequently every
 143 overlap discrepancy is bounded by

$$\delta_r = L\sqrt{2} N\sqrt{1 - r^2}. \quad (6)$$

144 Here discrepancy is the norm of the difference. At $r = 1$, the observations
 145 agree exactly. The factor N makes this estimate potentially weak in large
 146 populations; it supplies a pointwise guarantee from a population average. A
 147 shared scalar encoder is a special case. Dropping the overlap relation permits
 148 disagreement even at perfect locking. A common nonexpansive linear projec-
 149 tion preserves agreement, but agreement of a projection does not establish
 150 agreement of the full vector.

151 Mix retained fields with observations at gain $\eta \in [0, 1]$: $s_i \mapsto (1 - \eta)s_i +$
 152 ηo_i . Content and observation discrepancies bounded by ε and δ give a next
 153 discrepancy at most $(1 - \eta)\varepsilon + \eta\delta$. Repeated observations bounded by δ
 154 therefore give

$$\varepsilon_n \leq (1 - \eta)^n \varepsilon_0 + \delta. \quad (7)$$

155 Thus agreeing observations preserve exact compatibility and, for $\eta > 0$, re-
 156 duce an initial mismatch. Approximate observations leave a nondecaying
 157 term in the error bound. Taking a uniform bound on Eq. (6) connects this
 158 update to the phase process.

159 **3.3 What approximate agreement determines**

160 Exact gluing requires exact restrictions. For nonnegative profiles on a finite
 161 substrate, suppose every overlap discrepancy is at most ε in uniform distance.
 162 Weights that vanish outside their patches and sum to one at each site select
 163 a global profile by weighted averaging. The selection lies within ε of each

164 patch there, and any two such selections differ by at most ε , independently
165 of cover multiplicity. An L -Lipschitz readout consequently varies by at most
166 $L\varepsilon$ across selections.

167 At zero tolerance, the selection is the unique exact extension. At every
168 positive tolerance there are families with no exact extension; selection then
169 gives an approximation whose dependence on weights is bounded. Supple-
170 mental Material gives a three-site witness attaining the bound. The mass-
171 profile construction uses the scalar case of the content dynamics and addi-
172 tionally requires nonnegativity of the initial profiles and observations. The
173 general vector result supplies no identification with the spatial-measure sheaf.

174 These constructions concern declared objects: a specified cover, contents
175 valued in a normed space, and encoders related on overlaps. They select
176 neither the neural cover nor the decoder. Identifying their contents with
177 perceptual content, and establishing that the identification bears on experi-
178 ence, is a further modelling task; Section 6 sets out what a measurement of
179 it would have to establish.

180 4 Self-representation: a state that models it- 181 self

182 4.1 The reflexive construction

183 Compatibility gives a global state. Reflexivity relates it to an encoding within
184 that state. Equip global sections with a complete metric. If the additional
185 restriction-resonance condition identifies encoding with restriction to a region
186 U , the reconstruction map is

$$F(s) = \text{readout}(s|_U). \quad (8)$$

187 The region and readout are fixed across the state space. If F contracts and
188 fixes the section s_* formed from the local descriptions, Banach's theorem
189 makes s_* its unique fixed point (Banach, 1922). The formal bridge E89
190 supplies the contraction and $F(s_*) = s_*$; restriction resonance is separate.
191 Supercritical coupling makes the stipulated factor $q = e^{-(K-2D)\tau/2}$ less than
192 one for $\tau > 0$. Constructing a map with that factor remains a modelling
193 obligation.

194 In the cup example, the global state includes the represented scene and the
195 perceiver's relation to it. A local encoding must retain enough information
196 for the readout to recover that modelled state; cup position alone would be
197 insufficient if hand configuration and viewpoint could not be reconstructed.

198 For the same map with Lipschitz factor $q < 1$, approximate reconstruction

199 gives

$$d(s, F(s)) \leq \varepsilon \implies d(s, s_*) \leq \frac{\varepsilon}{1-q}. \quad (9)$$

200 At the stipulated rate this radius grows as $2\varepsilon/((K-2D)\tau)$ near threshold, for
 201 fixed positive ε and τ . Exact uniqueness survives while the error guarantee
 202 weakens.

203 4.2 Reconstruction across a family of states

204 Sensitivity can be required quantitatively. Declare relevant macrostates, an
 205 encoder E and readout R , and require $d(s, R(E(s))) \leq \varepsilon$ throughout that
 206 family. If two states have the same code, the triangle inequality gives

$$E(s) = E(t) \implies \max\{d(s, R(E(s))), d(t, R(E(t)))\} \geq \frac{1}{2}d(s, t). \quad (10)$$

207 Thus M relevant states pairwise more than 2ε apart require at least M
 208 distinguishable codes; an alphabet of at most 2^b codes requires $M \leq 2^b$.
 209 Applied to the reflexive boundary, E is its local encoding and $R \circ E$ is the
 210 existing self-prediction map.

211 This is an additional accuracy criterion on a family, with no requirement
 212 for multiple exact fixed points. A blind encoder or constant readout fails it
 213 for separated states despite having a unique fixed point.

214 4.3 Agreement after a change takes communication

215 Snapshot compatibility does not specify how quickly descriptions respond
216 when a shared quantity changes. In a synchronous network whose updates
217 read only the site’s own state and declared neighbours’ messages, two initial
218 configurations agreeing on a site’s causal past through T rounds produce the
219 same state there at round T . If two fresh interventions outside that past
220 require reports separated by more than 2ε , no report from that site can be
221 within ε of both. This follows from the reconstruction bound in Section 4.2.

222 A three-site delay line witnesses failure after one round and exact report-
223 ing after two; cutting the path prevents delivery at every round. A report
224 may still happen to be correct about one fixed world. The bound concerns
225 responsiveness across interventions.

226 Throughout this section *prediction* names a reconstruction map; reading
227 it as temporal neural prediction would require dynamics. The relevant family
228 of states, the metric, the tolerance, the communication graph and the dead-
229 line are all declared inputs, as are the represented variables and the fidelity
230 counted as accurate. Reading the resulting code count as physical memory
231 capacity requires a hardware model, and a readout that varies with the state
232 requires a further one.

233 5 Thermodynamics and composition

234 The phase and representation results concern coupling, content disagreement
235 and reconstruction error. Thermodynamics constrains the physical opera-
236 tions that could implement them. Finite-register capacity obeys $H(p) \leq$
237 $\log |X|$; erasure heat depends on specified system–bath assumptions (Lan-
238 dauer, 1961; Norton, 2005; Shenker, 2000). A separate boundary result con-
239 cerns a field reaching disconnected vacuum components (Kibble, 1976). The
240 composition connects these objects through the explicit hypotheses in Ta-
241 ble 1.

242 5.1 Feedback changes the information constraint

243 For passive memory X_t driven by an environmental signal S_t , the predictive
244 bound is

$$W_{\text{diss}} \geq k_B T [I(X_t; S_t) - I(X_t; S_{t+1})]. \quad (11)$$

245 The Markov structure $X_t \rightarrow S_t \rightarrow S_{t+1}$ makes the bracket nonnegative (Still
246 et al., 2012). Zero retained information also satisfies it, so a task and a
247 learning rule require additional specification. Relating this bound to Lan-
248 dauer heat requires a common thermal scale and budget, and memory waste
249 bounded by erased entropy.

250 An acting agent changes the signal it will receive. Specify policy, world,

251 observation and memory-update channels. For their finite laws,

$$\begin{aligned}
I(X_t; S_t) - I(X_t; S_{t+1}) &= I(X_t; S_t \mid S_{t+1}) \\
&\quad - I(X_t; S_{t+1} \mid S_t).
\end{aligned}
\tag{12}$$

252 The second conditional term vanishes in the passive model. With feedback,
253 the signed difference can be negative as action creates correlations with the
254 future. A causal reading requires an environmental state sufficient to describe
255 the relevant dynamics.

256 For a controlled substep, the active thermodynamic bound instead uses
257 the joint entropy. Forward and reverse path laws give $\Sigma = D_{KL}(P_F \parallel P_R) =$
258 $\Delta H(X, S) + \langle Q \rangle / \theta \geq 0$ under local detailed balance, with $\theta = k_B T > 0$ and
259 heat positive into the reservoir (Horowitz and Esposito, 2014; Ito and Sagawa,
260 2013). Every forward path of positive probability must have a positive-
261 probability reverse for this finite identity. A missing reverse gives infinite
262 divergence. The supplement develops the support conditions and composes
263 actuation with memory updates on their shared intermediate law.

264 **5.2 Learning and repeated operation need explicit re-** 265 **sources**

266 The finite examples distinguish selecting a feasible policy, relaxing a register
267 toward an encoded reward, and learning an unknown task from observa-
268 tions. The observational learner acquires correlation with an environmental

parameter while its action marginal stays uniform. A continuing model executes actuation, observation-driven learning and reset in sequence, preserving above-chance performance at a positive work cost per cycle.

A phase-derived sensor makes the informational distinction concrete: an informative readout and a blind control have the same phase order but different learning performance. Attaching the learner’s register to the actuator of Section 2.2 ties reward, installed kernel and work to its own executed law. The learning witness rises above the necessary installed-energy bound while its matched control remains on the exclusion line. These finite constructions, detailed in Supplemental Material, leave the evolving feedback between phase dynamics, installation and sensing to be specified.

For that model, common energy E , thermal scale θ , local detailed balance and positive support at each stage give

$$\theta[H(p_0) - H(p_N)] \leq Q_{<N} = W_{<N} + \langle E \rangle_0 - \langle E \rangle_N \leq B + \langle E \rangle_0 - \langle E \rangle_N, \quad (13)$$

where expected cumulative work stays within the supplied allowance B at every prefix. This bounds expectations. Separate constructions track a store on each trajectory, include finite sources, charge sensor clearing and prepend preparation. They specify when a mean allowance represents an available resource; an expected budget alone permits overdrawn paths.

These constructions establish costs for their declared processes. A finite source with a positive net cost permits only finitely many funded cycles,

289 while continuing operation requires replenishment. Register–bath witnesses
 290 also distinguish actual bath-energy transfer from transition-log heat. Identifying a work store, an erasure budget and the installation energy of Section 2.2 requires physical relations that these separate models do not supply
 291 automatically.
 292
 293

294 5.3 What the composition establishes

295 The passive theorem `chain` and feedback theorem `chain_active` each accept eight connecting implications (Table 1). In the active branch, a named
 296 substep at the register’s thermal scale satisfies
 297

$$\theta > 0, \quad \theta[H(p) - H(p')] \leq \langle Q \rangle \leq B, \quad (14)$$

298 with B the register’s heat. The upper allocation follows from a ledger whose
 299 operations deliver that total heat to one reservoir and whose other operations
 300 do not increase joint entropy. Those ledger assumptions are physical inputs.
 301 Neither Landauer’s lower bound nor microscopic reversibility supplies them.

Table 1: Connecting hypotheses, eight per branch, and the installed-energy consequence. The branches share E12, E23 and E56–E89. Exact overlap compatibility and the encoding–readout structure are additional inputs. Supplemental Material, Table S1, maps the node-level results.

Connection	Status	Required assumption
E12	Physical premise	A finite-capacity system realizes a field leaving its vacuum manifold. Capacity and boundary are independent roots.

Table 1 (continued)

Connection	Status	Required assumption
E23	Bridge assumption	The boundary is associated with a finite register whose update leaves states unreachable.
E34 / E34Active	Bridge assumption	Passive: predictive memory at the register's thermal scale and heat budget, with nonzero waste. Active: positive paths, local detailed balance and allocation of register heat to the named substep; the allocation uses the ledger identity and other operations' compressiveness.
E45 / E45Active	Modelling assumption	Convergent spatial coupling energies. Declared mode hardware supplies a kernel limit for either branch. A learning witness computes installation and work from its own executed law. Refinement and physical identifications remain inputs.
E56	Physical commitment	A specified physical field realizes the limiting coupling and satisfies the kernel conditions.
E67	Physical commitment	That field operates above the synchronization threshold.
E78	Bridge assumption	Coherence relaxes the supplied cover at coupling at least the named mean-field value. Shared phase encoders bound vector-content mismatch; the exact cover and timely responsiveness remain separate requirements.
E89	Modelling assumption	A map mediated by local self-encoding fixes the glued section and has the stipulated Lipschitz factor $q < 1$. Reconstruction over a declared family imposes an additional code-separation criterion.
Installed energy	Conditional theorem	Under Section 2.2's occupancy and mode-price conditions, E56 and E67 require $U_{\text{inst}} > 2D/\kappa$. The learning witness clears this necessary bound; its matched control does not. Neither physical premise follows.

303 lap compatibility and a reflexive boundary specifying encoding and readout.
 304 They conclude `UnifiedSelf`: the glued section is the unique fixed point of
 305 self-prediction. Compatibility and accuracy at that section enter as premises;
 306 uniqueness follows. Identifying encoding with restriction needs the separate
 307 resonance condition of Eq. (8).

308 Joint witnesses combine a register, boundary, refining mesh and small
 309 spatial cover; active witnesses add controlled thermal updates and a priced
 310 installation. They establish satisfiability under their stated identifications.
 311 Mesh convergence follows from refinement data, and the witnesses establish
 312 no common cortical mechanism. Supplemental Material maps the claims to
 313 their Lean declarations and records the counterexamples. The development
 314 declares no additional axioms beyond its libraries' foundational axioms.

315 6 What the observations establish

316 The composition holds under its premises. Whether observation can estab-
 317 lish those premises is a separate question, and the three requirements it
 318 connects concern different objects: phase dynamics, local descriptions and
 319 internal encodings. Their observables must retain these distinctions. The
 320 following controls show which mechanisms remain indistinguishable, which
 321 temporal approximations fail, and what independent content measurements
 322 must establish.

323 6.1 Stationary distributions leave the mechanism un- 324 determined

325 Equation (2) gives a concentration–coherence relation with no fitted shape
326 parameter. Estimate r from circular resultant length and a from density
327 shape: inverting R from the measured r would make agreement automatic.
328 Even independently estimated agreement identifies a statistical family shared
329 by different mechanisms.

330 Let $q(t)$ cross 2, let $r(q)$ be its stationary branch, and fix $D_0 > 0$. Increas-
331 ing coupling $K(t) = D_0 q(t)$ at diffusion D_0 and decreasing diffusion $D(t) =$
332 $K_0/q(t)$ at coupling K_0 both give concentration $a = q(t)r(q(t))$. Uncoupled
333 oscillators with prescribed drift $-h(t) \sin(\theta - \psi)$, where $h(t) = D_0 q(t)r(q(t))$,
334 and diffusion D_0 have that same density. Their stationary distributions co-
335 incide window by window despite different causes.

336 Resolved dynamics supply additional information. In the ideal model with
337 continuous paths, quadratic variation accumulates at rate $2D$ and conditional
338 increment means constrain drift. Separating endogenous coupling from a
339 matched common drive additionally requires a measured drive or controlled
340 perturbation; identifying electromagnetic coupling requires evidence specific
341 to that interaction. Supplemental Material gives the equivalence construction
342 and its numerical check.

343 **6.2 Finite ramps change the observable onset**

344 The square-root law in Eq. (3) describes stationary solutions. A changing
345 coupling requires time for the phase distribution to approach those solutions.
346 In a deterministic mean-field control that crosses the threshold symmetrically,
347 none of the scanned rates meets the declared tracking tolerance: slowing
348 the crossing does not remove the terminal residual under that initialization.
349 A slow control variable therefore supplies no general guarantee of branch
350 tracking at onset. Supplemental Material gives the residual, rate range and
351 admissible rates away from threshold.

352 The finite-population ramp study uses $N = 2000$, $D = 1$ and 32 replicas
353 at four coupling speeds. Escape is the first of three consecutive saved samples
354 at $r \geq 0.2$ after threshold. Faster ramps delay escape (Fig. 3); only 5/32
355 replicas escape within the fastest window. The three slower ramps have
356 mean coupling delays 0.1947, 0.0684 and 0.0253. Fitted onset exponents
357 differ from those of the exact stationary branch evaluated on the same grids.
358 At the fastest speed the ensemble mean never reaches the fit window, so its
359 fitted slope does not estimate stationary onset.

360 Delayed passage through a noisy bifurcation (Berglund & Gentz, 2002)
361 makes the apparent exponent depend on ramp speed and fitting window.
362 The specified deviation-based critical ramp speed is not bracketed by this
363 sweep. With fluctuating coupling, the relevant stationary comparison also
364 depends on rate: the mean coupling approximates order in the tested fast
365 regime, while averaging stationary order over coupling works for the tested

slow drives confined to one side of threshold. Slow crossing drives can satisfy
neither approximation. These are finite-horizon observations, with controls
and parameters in Supplemental Material.

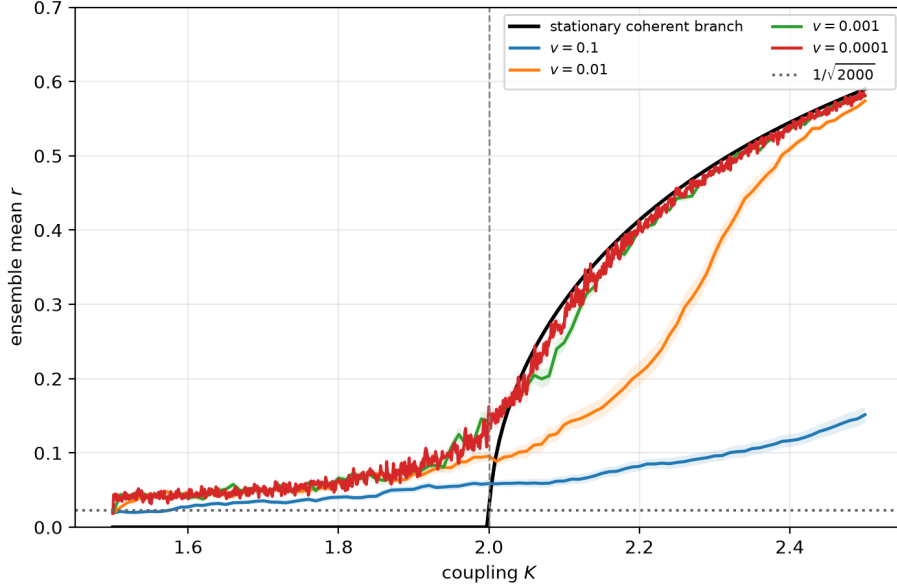


Figure 3: Finite-rate departures from the stationary branch. Ensemble-mean order versus coupling for four ramp speeds v , with one-standard-error bands across replicas. The dashed vertical line is the static threshold and the dotted horizontal line the finite-size reference. Faster ramps delay escape, changing the apparent recovery onset.

Additional controls address integration bias, population size and spatial
or signed coupling. The refined signed-network crossing bracket $K_{\text{eff}}/D \in$
 $[1.958, 2.009]$ contains 2. These are finite numerical regimes, with full pa-
rameters and controls in Supplemental Material; they supply no cortical cal-
ibration.

374 6.3 A winding state leaves the order parameter am- 375 biguous

376 The order parameter r is indifferent to where a phase sits in space, so a
377 field winding once across a periodic sheet has $r = 0$ while every patch of
378 it stays locked. Such a field is not an artificial construction here: on a
379 torus a twist is stationary for any isotropic symmetric kernel at identical
380 frequencies, because the coupling term is odd in the separation against a
381 kernel even in it, which `char_is_kuramoto_trajectory` proves for any finite
382 periodic array, this sheet included. It satisfies the positivity and identical-
383 frequency hypotheses, and violates only the quarter-turn initial spread that
384 the convergence result additionally requires before placing the limit at the
385 uniform state.

386 On the sheet of Section 2's spatial control, a twist survives below an
387 interpolated decay length of 0.1871 mm and unwinds above it, a boundary
388 interior to the 0.1–0.3 mm band. At 0.1 mm the same sheet at the same
389 coupling, diffusion and kernel reaches $r = 0.9657$ from a uniform start and
390 $r = 0.0083$ from a twisted one, both stationary, with patch-local order 0.9015
391 and no winding defect. At fixed K the observable therefore takes two values
392 spanning nearly its whole range: K does not determine r , and a low resultant
393 does not by itself distinguish weak coupling from a winding field.

394 Under the quenched frequency spread of 1.5 rad s^{-1} that the spatial con-
395 trol carries, the boundary moves to 0.1421 mm and the same comparison

396 at 0.1 mm gives 0.9418 against 0.0348, holding across 3 frozen realisations.
 397 Heterogeneity shortens the range that sustains a wave without removing it.
 398 This qualifies what a phase trace can establish. An onset criterion stated
 399 in r alone registers a change of basin as readily as a change of coupling, so
 400 the protocol of Section 7.3 needs a spatial observable beside the resultant.
 401 Supplemental Material gives the sweep, its uniform control and the limits of
 402 both.

403 **6.4 An amplitude layer changes the state space, not** 404 **the reading**

405 A phase state carries no amplitude: every oscillator sits on the unit circle,
 406 so a field with an amplitude node has no representation in these dynam-
 407 ics. The real Stuart–Landau extension supplies one on the same coupling,
 408 leaving intact the gradient structure the convergence results rest on. The
 409 state becomes $z_i \in \mathbb{C}$ with $\dot{z}_i = (\mu - |z_i|^2)z_i + \sum_j A_{ij}(z_j - z_i)$, and its phase
 410 equation is the Kuramoto equation with each coupling weighted by a ratio
 411 of amplitudes (`amplitude_phase_is_kuramoto`), so the phase model is this
 412 one with the amplitude ratios set to one rather than a separate object.

413 What the added degree of freedom decides is which windings exist at all.
 414 On a periodic array of any dimension, the sheet of this section included, a
 415 winding requires $\mu + \lambda > 0$, where $\lambda = \sum_d f_d(\cos \phi_d - 1)$ is the eigenvalue
 416 the kernel supplies, summed over separations with ϕ_d the phase the winding

417 advances across d (`exists_charState_iff`; on the ring $\phi_d = 2\pi qd/n$). For
 418 the nonnegative kernels used here $\lambda \leq 0$, and moving coupling mass to longer
 419 separations lowers it further, so a broad enough kernel removes the winding
 420 outright. That is the direction the sweep above measures, reached by a
 421 different route: the phase model retains every winding at every coupling and
 422 loses this one to instability instead. The two layers therefore bound the same
 423 regime from opposite sides, and a phase recording does not separate a state
 424 that has ceased to exist from one that has ceased to be stable.

425 These states are windings and not spirals, and on the sheet that is a
 426 theorem rather than a shortage of dimensions. A cortical phase singularity
 427 is a point at which the amplitude vanishes and around which the phase turns
 428 once, read in practice as the circulation around a small plaquette; the states
 429 above have modulus equal at every site, so that circulation is exactly zero
 430 on every plaquette (`plaquetteCirculation_eq_zero_of_winding`). Nor can
 431 the layer host one defect: over a periodic sheet the circulations of *any* phase
 432 field sum to zero, so a defect cannot stand alone — they arrive in cancelling
 433 pairs (`no_isolated_defect`). The sweep’s near-zero measured defect density
 434 agrees with the first of these and says nothing beyond it. Supplemental
 435 Material gives the layer, the standing wave it admits inside the band, and its
 436 remaining limits, of which the one a reader of the neuroscience meets first is
 437 that relaxational dynamics has no propagation speed, the quantity a cortical
 438 wave measurement leads with.

439 **6.5 Coherent adaptation can lose environmental align-** 440 **ment**

441 An adaptive coupling rule might supply representational structure, but its
442 objective must reward the relevant agreement. For symmetric coupling, pair-
443 wise sine terms cancel in the population mean drift. Minimizing squared drift
444 at that fixed mean therefore rewards equalizing drift across nodes. In the
445 frequency-cluster control, this favors coupling between unlike-frequency clus-
446 ters, whereas the diagnostic environmental template rewards coupling within
447 clusters.

448 The formalization proves descent of the squared-drift objective for a spec-
449 ified gradient flow at fixed phases. Its functional differs from thermodynamic
450 entropy production: at a coherent stationary density the probability current
451 vanishes while phase-averaged squared drift is positive. Supplemental Mate-
452 rial gives the operator, its gradient and the conditions of the descent result.

453 The finite control co-evolves noisy phases and symmetric, nonnegative
454 couplings at fixed total resource. Its template is a diagnostic covariance of
455 latent signals, never a training input. Across three seeds, second-half order
456 remains between 0.9167 and 0.9222, while the within-cluster to between-
457 cluster coupling ratio falls from 0.9824–0.9948 to 0.3507–0.6135 (Fig. 4).
458 Random-update, relabelled-gradient and frozen-kernel controls are reported
459 in Supplemental Material.

460 The immediate post-update objective reduction disappears when aver-

461 aged over complete update intervals. A prespecified grid of 6 rate–cadence
462 combinations contains no case meeting the 5% sustained-reduction criterion
463 while retaining coherence and losing alignment. The finding is coherent adap-
464 tation away from the specified structure, with no demonstrated sustained cost
465 reduction.

466 Kernel alignment and overlap compatibility are different observables.
467 This control addresses acquisition of its designated template under its par-
468 ticular objective. Learning models can explicitly include data in an objective
469 (Corberi et al., 2026); an account of agreement about the cup must specify
470 the content its learning rule is meant to preserve.

471 **6.6 Content agreement needs independent decoders**

472 A neural test needs independently identified domains, local descriptions and
473 restriction maps. Two decoders should estimate a shared content on overlap-
474 ping domains before their agreement is examined. Discrepancy requires a null
475 under decoding noise and controls using deliberately compatible and incom-
476 patible examples. An implementation on constructed data separates these
477 cases at $AUC = 0.990$. Shared-prior shrinkage can nevertheless manufacture
478 agreement, and regional bias can manufacture disagreement. Per-patch de-
479 coding diagnostics detect these failures; the compatibility statistic alone does
480 not.

481 Phase agreement cannot replace that measurement. On constructed con-
482 tent, phase carries a share 1.000 of the content entropy when content is built

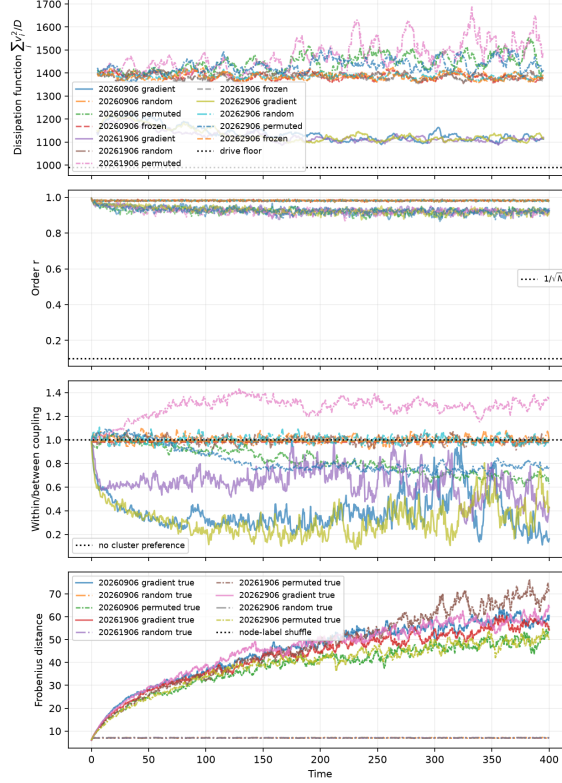


Figure 4: Coherence persists while coupling loses alignment with the frequency-cluster template. Gradient updates (solid), random updates (dash-dotted), relabelled gradient updates (dash-dot-dotted), and a frozen kernel (dashed) share one phase-noise stream per seed. Panels show the immediate post-update squared-drift objective, phase order, within-cluster to between-cluster coupling, and template distances. The gradient arm preserves order while its coupling ratio falls; growth in both distances mainly reflects kernel norm. The objective panel samples the instant after updates; complete-interval measurements find no sustained reduction in the bounded study.

483 from it and 0.000 when it is not. A phase-derived statistic reports com-
484 patibility on phase-locked patches whose local descriptions disagree. The
485 implemented content statistic detects their incompatibility. These controls
486 validate a measurement on known constructions; a neural content model and
487 decoder remain to be established.

488 Choosing the cover also chooses a candidate subject: trivial descriptions
489 can agree while carrying little content, and a partition can conceal disagree-
490 ment. The useful choice must therefore be constrained by what neural ac-
491 tivity encodes. The exact composition assumes compatibility of its supplied
492 cover. The content mechanisms above identify additional hypotheses under
493 which agreement can arise; they leave the empirical choice of content and
494 the connection to that cover explicit.

495 **6.7 Phase-density estimation and measurement feasi-** 496 **bility**

497 Public propofol-sedation EEG (Bajwa et al., 2025) exercises the estimators
498 by pooling channels in 100-ms windows. Bipolar and common-reference mon-
499 itages produce different residuals. The bipolar results are compatible with the
500 Bessel relation, but their concentration range cannot distinguish it from its
501 linear approximation at the observed scatter. Temporal pooling changes the
502 estimand, and sedation recordings leave the awakening hypothesis untested.

503 The concentration range and dependence determine whether the relation

504 can be tested. Under the specified log-density estimator and bias-floor crite-
505 rion, discrimination at the observed concentration ceiling $a = 0.542$ requires
506 3000 independent sites in the scanned designs, rising to 31000 when sites
507 share a mean in groups of 10. These counts describe that estimator and de-
508 pendence model; the intended recording requires its own calibration. Supple-
509 mental Material reports the bin-count controls and the full dependence-aware
510 design.

511 An informative dataset needs sufficient concentration range or precision,
512 calibrated estimators and an observation model for dependence, source mix-
513 ing and reference choice. Public mouse calcium imaging provides thousands
514 of spatially located neurons across waking, sleep and anesthesia (Oomoto et
515 al., 2026), but its 7.65 Hz frame rate is below the phase-estimation band used
516 here. Site count, state coverage and bandwidth are separate requirements.

517 **7 The cortical hypothesis and an awakening** 518 **test**

519 The cortical proposal identifies a physical source of coupling and asks whether
520 its changes help explain recovery. The tests in Section 6 determine what
521 observations must distinguish before a phase trace can support this identifi-
522 cation.

523 7.1 Physical identification and calibration

524 The cortical hypothesis identifies part of effective coupling with endogenous
525 extracellular electromagnetic interactions. Fields can modulate spike timing
526 (Fröhlich & McCormick, 2010; Anastassiou & Koch, 2015), and field models
527 propose coordination of distributed neural information (Pinotsis & Miller,
528 2022; McFadden, 2020). Phase organization under propofol motivates study-
529 ing this regime (Bastos et al., 2021), but does not measure coupling.

530 The formal field predicate requires a jointly continuous kernel, a normal-
531 ized substrate, coupling equal to the kernel average, and no single site pro-
532 viding more than half that strength. Toy constructions satisfy it. A cortical
533 realization must specify the scale at which the kernel and its reduction ap-
534 ply: calcium imaging reports different state-dependent modularity patterns
535 at single-cell and mesoscale resolution (Kiyooka et al., 2026).

536 The spatial reduction is also testable. A row-sum aggregation at zero
537 diagonal tracks the uniform-coupling control over the tested spatial decay
538 range, while frequency heterogeneity displaces the operational threshold. The
539 fit window biases both estimates relative to $2D$; agreement with the control
540 is therefore the supported conclusion. Supplemental Material reports the
541 reduction, its numerical error and the distinct assumptions needed for the
542 kernel-average identification in Eq. (5).

543 The field commitment requires physical evidence beyond a choice of math-
544 ematical representation. A finite matrix spread over positive-mass cells pre-
545 serves its total coupling and phase correlation exactly. Point-supported

546 vanishing under an atomless measure and a disadvantage from excluding
 547 a better-correlated pair concern specified support constraints. Neither com-
 548 parison supplies energy prices or a biological–digital distinction.

549 A spike-time shift alone supplies no coupling rate. The dimensionless
 550 product of population size N , timing sensitivity s , field amplitude E and
 551 frequency f needs a conversion rate γ :

$$K_{\text{eff}} = \gamma N s E f. \quad (15)$$

552 This rate describes how induced timing shifts accumulate into drift. Nei-
 553 ther it nor the installation factor κ follows from a heat bound. Testing the
 554 physical threshold requires independent estimates of the interaction kernel,
 555 its aggregation and diffusion in the same cortical regime.

556 **7.2 Extracellular geometry and recovery**

557 The proposed control variable is extracellular geometry. Extracellular vol-
 558 ume fraction is approximately $\alpha = 0.2$ at tortuosity $\lambda \approx 1.6$ (Syková &
 559 Nicholson, 2008); a sleep study reports a rise from roughly 0.14 to 0.23 (Xie
 560 et al., 2013). Below 10 kHz, cortical tissue conducts approximately resistively
 561 at $0.3\text{--}0.6\text{ S m}^{-1}$ (Barbour, 2017). At fixed transmembrane current density,
 562 an effective-medium description predicts that a larger extracellular fraction
 563 raises conductivity and reduces field amplitude.

564 The coupling-gated hypothesis consequently predicts that extracellular

565 contraction on awakening raises E and, holding the other factors in Eq. (15)
 566 fixed, raises effective coupling. This argument supplies a direction; the
 567 geometry–coupling relation, its magnitude and its role in human recovery
 568 require measurement.

569 For a smooth crossing at nonzero speed, evaluating the stationary branch
 570 along the changing control gives the benchmark

$$r_{\text{ss}}(K(t)) \sim \sqrt{\frac{\dot{K}(t_0)}{D}} (t - t_0)^{1/2}, \quad K(t_0) = K_c, \quad \dot{K}(t_0) > 0. \quad (16)$$

571 This follows from Eq. (3) at fixed D and concerns the stationary branch
 572 evaluated at $K(t)$. Reading it as observed recovery requires an indepen-
 573 dently justified tracking approximation. The crossing controls in Section 6.2
 574 leave that approximation unsupported at onset, even though a separation of
 575 geometry and phase time scales can support it away from threshold. The
 576 awakening hypothesis therefore needs a calibrated dynamical prediction for
 577 an operational onset above the finite-size fluctuation floor. Crossing K_c alone
 578 does not supply a recovery latency.

579 **7.3 A protocol for the awakening experiment**

580 The physical test has six requirements:

- 581 1. **Coupling and diffusion in the same units.** Estimate them from re-
 582 solved phase dynamics, separating heterogeneity and observation noise.
 583 Measure the drive or perturb the system to distinguish endogenous cou-

584 pling.

585 **2. Geometry calibrated to coupling.** Validate the geometry observ-
586 able and its relation to effective coupling, with uncertainty. The con-
587 ductivity argument supplies only the expected direction.

588 **3. Calibrated phase measurements.** Estimate r by resultant length
589 and a by unweighted regression of log density on centered cosine.
590 Choose site count and binning from a dependence-aware calibration
591 over the intended concentration range. The tested designs require
592 thousands of sites at the observed concentrations; temporal pooling
593 changes the spatial estimand.

594 **4. An independent recovery endpoint.** Measure behavioral recovery
595 separately from phase organization.

596 **5. A predeclared dynamical onset criterion.** Set a coherence thresh-
597 old above the finite-size floor and evaluate its latency under the mea-
598 sured coupling trajectory, including ramp speed or fluctuations. Record
599 a spatial observable beside the resultant, since a winding field reports
600 a low resultant at coupling that sustains local order (Section 6.3).

601 **6. Matched model comparison.** Compare coupling increase, diffusion
602 change, common drive and alternative recovery models under the same
603 observation and filtering assumptions.

604 Failure of the concentration–coherence relation over an informative range
605 constrains the density model. Failure of the calibrated geometry–coupling
606 association constrains the physical gate. Better prediction of independent
607 recovery by a competing mechanism counts against its proposed role. A
608 favorable outcome still leaves the content of coordinated activity to be ex-
609 plained.

610 8 Discussion

611 Stated positively, and granting the premises collected in Table 1, the account
612 runs as follows. Extracellular contraction on awakening raises endogenous
613 field amplitude and with it the effective coupling between cortical oscillators.
614 Once that coupling passes $2D$, a coherent phase branch exists and the popu-
615 lation settles onto it. Encoders shared where cortical territories overlap turn
616 that phase order into agreement about the quantities the territories describe
617 in common, so the local descriptions glue: there is one state of which each
618 is a restriction. Part of that state encodes the state itself, and a readout of
619 that part reconstructs it accurately enough, and sensitively enough across
620 the states that matter, to make it the unique state the system models itself
621 as being. On this account the glued state is what makes distributed activity
622 one situation, and its accurate self-reconstruction is what gives that situation
623 a perspective.

624 Every step of that description is conditional, and its conditions are the

625 subject of what follows.

626 **8.1 Unity and a minimal self as empirical commit-** 627 **ments**

628 We propose the state glued from compatible local descriptions as a corre-
629 late of conscious unity, and its accurate internal reconstruction as a correlate
630 of a minimal self. The motivations are that different contents belong to
631 one experienced situation and that the situation is experienced from a per-
632 spective. The conditional results identify relations a realization would need:
633 sufficient installed resources for the specified phase model, agreement about
634 shared content, and an encoding that supports reconstruction at the required
635 fidelity.

636 Compatibility allows different descriptions to contribute different infor-
637 mation. It also permits uncertainty and conflicting judgments: apparent cup
638 position and judged cup position can be distinct quantities in a consistent
639 description of their discrepancy. Which quantities are shared must be de-
640 termined independently. Choosing them solely to secure agreement would
641 deprive the proposal of empirical force.

642 The minimal-self proposal concerns the perceiver’s position and condition
643 within the situation, such as the cup’s location relative to a body that can
644 reach it. The encoding could be implicit and need carry no autobiographi-
645 cal narrative. Its proposed necessity is philosophical: even an unreflective

646 episode has a perspective. Gluing by itself requires no self-encoding, and re-
647 constructability alone supplies no functional role. Altering the encoded hand
648 or viewpoint while holding the external scene fixed should change estimates
649 of the perceiver’s relation to the cup as the readout predicts. Whether that
650 represented perspective is experienced remains a further question.

651 **8.2 What a test of the full proposal must establish**

652 Both uniqueness results concern specified objects. They choose neither the
653 cover nor the encoding, and do not select between candidate subjects. In-
654 tegrated information theory treats exclusion explicitly (Tononi et al., 2016);
655 here selection requires independent evidence about neural organization. A
656 constant readout can contract and fix a predetermined state while discard-
657 ing its input. The family criterion in Section 4.2 tests the sensitivity that
658 this construction lacks, provided the relevant states and fidelity are declared
659 independently.

660 A test must separately measure coupling relative to diffusion, content
661 compatibility and self-reconstruction, alongside an independent endpoint for
662 experience. Meeting all three without a unified episode would challenge suffi-
663 ciency. A unified episode with incompatible descriptions would challenge the
664 unity identification; one lacking the specified self-encoding would challenge
665 the reflexive requirement. Awakening, dreaming and dissociative states offer
666 possible settings, but absence of report cannot by itself define absence of
667 experience.

668 The next physical requirement is a calibrated geometry–coupling relation
 669 and resolved phase dynamics. The installation bound contributes a nec-
 670 essary resource condition where its hardware representation applies; cortical
 671 mode prices and field-mediated coupling require measurement. For the repre-
 672 sentational connection, an informative experiment asks whether a controlled
 673 change in coordination alters independently decoded content agreement, with
 674 controls for sensory information and decoder quality. The constructed-data
 675 estimator establishes feasibility for specified contents. Neural local descrip-
 676 tions, shared overlap encoders and the self-encoding have yet to be identified.
 677 The population-size dependence of the phase-to-content bound further limits
 678 its quantitative use.

679 The separation results explain why these measurements must be inde-
 680 pendent. A stationary distribution leaves several recovery mechanisms indis-
 681 tinguishable. Equal phase order can accompany incompatible descriptions
 682 or different task information. The tested plasticity rule preserves coherence
 683 while losing its designated template. Each finding concerns the specified
 684 model and observable; their role is to delimit the evidence needed for the
 685 proposed connections.

686 **8.3 GPUs and the limits of no-go claims**

687 For a large language model (LLM) running on graphics processing units
 688 (GPUs), the framework requires identifying a physical system whose local
 689 descriptions form one state and whose internal encoding reconstructs that

690 same state. The coupling-gated route E56–E78 requires a calibrated physi-
691 cal coupling, its noise regime and a connection from coordination to content
692 agreement. An attention matrix or a shared clock supplies none of these iden-
693 tifications on its own. Computing a field model likewise leaves its realization
694 in the computing substrate to be established. These physical obligations also
695 apply to the cortical proposal.

696 Unity requires the candidate’s local descriptions to agree on indepen-
697 dently identified shared quantities (Section 3). Common timing and fluent
698 output do not establish that agreement. Digital representations can sat-
699 isfy compatibility, however, so finite hardware is compatible with the gluing
700 construction. The empirical commitment is to identify the relevant cover
701 and contents with one experienced situation; mathematical uniqueness alone
702 selects no subject.

703 For self, the system must carry an encoding $E(s)$ and an internal readout
704 R of the same global state. E89 requires $F = R \circ E$ to contract and fix
705 the state produced by gluing. A constant readout shows why a unique fixed
706 point alone is insufficient for informative self-representation. The additional
707 family criterion in Section 4.2 supplies a test: if two relevant macrostates
708 satisfy $E(s) = E(t)$ and $d(s, t) > 2\varepsilon$, Eq. (10) rules out reconstructing both
709 within error ε . An intervention outside the encoding region’s causal past
710 gives a corresponding obstruction before a declared communication deadline.
711 Identical verbal reports alone do not establish identical internal codes. These
712 conditional no-go results exclude a specified reconstruction at a specified

713 fidelity or deadline, and apply to biological and digital candidates alike.

714 The hardware comparisons in Supplemental Material (“Coupling support
715 and continuum observables”) have similarly restricted scope. Fixed wiring
716 can impose a correlation disadvantage when it excludes a better-correlated
717 pair, but wiring both directions of a perfectly correlated pair permits an opti-
718 mal allocation. Point-supported kernels vanish under an atomless continuum
719 measure, whereas finite positive-mass cell representations preserve discrete
720 coupling weight and correlation exactly. Neither comparison makes the rep-
721 resentation conditions unattainable on GPUs. A universal exclusion would
722 require a necessary condition for consciousness and a proof that every rel-
723 evant GPU implementation violates it; the present conditional composition
724 supplies neither.

725 Whether experience requires a living substrate is the explicit subject of
726 biological naturalism (Seth, 2026a,b), on which the properties that ground
727 experience are those of self-sustaining organisms and are therefore not se-
728 cured by simulating them, while a sufficiently lifelike artificial system is not
729 excluded on grounds of material alone. The conditions stated here name no
730 biological mechanism, and computation described as such does not satisfy
731 them either: a candidate must exhibit a physically realized coupling in a mea-
732 sured noise regime, agreement about independently identified shared quan-
733 tities, and an internal reconstruction meeting a declared fidelity. Whether
734 any realization of those conditions that is not alive would be accompanied
735 by experience is the point at issue, and the present results do not settle it.

736 The neuroscientific case against *human-like* consciousness in the text-only
737 LLMs assessed by Aru and colleagues concerns embodied sensory interaction,
738 thalamocortical organization and biological regulation linking perception and
739 action to survival (Aru et al., 2023). Their absence supports skepticism con-
740 ditional on their causal relevance to experience; those biological mechanisms
741 are not requirements of the abstract gluing and reconstruction definitions.
742 Workspace-like representations reported in some LLMs (Gurnee et al., 2026)
743 provide candidate organization to investigate. Within this framework, the
744 relevant evidence would identify compatible contents and an informative in-
745 ternal reconstruction, or demonstrate their failure under controlled interven-
746 tions.

747 8.4 Explanatory scope

748 Integrated-information, global-workspace and predictive-processing accounts
749 organize unity, access and modelling differently (Tononi et al., 2016; Baars,
750 1997; Dehaene et al., 2014; Friston, 2010). The present proposal contributes a
751 candidate physical interaction and conditional relations between local content
752 and self-encoding. Gluing and reconstruction permit multiple substrates;
753 the electromagnetic commitment enters through physical calibration. The
754 formalization checks implications under their hypotheses, and its witnesses
755 establish realizability on declared models. They do not together supply a
756 cortical agent.

757 The explanatory question is whether the specified relations occur in neural

758 activity and matter for experience. In the cup example, this means establish-
759 ing how coordination helps maintain one represented situation and how the
760 perceiver’s place within it contributes to the episode. The proposal identi-
761 fies candidate correlates and tests of that identification. Why those relations
762 would be accompanied by experience remains unexplained.

763 **Data and code availability**

764 The Lean development, simulation scripts, tests and saved numerical sum-
765 maries are maintained in the GitHub repository `bobaseb/coupling_gated_`
766 `cortical_coherence`. Publication simulation values are generated from
767 saved summaries without rerunning production sweeps. The exploratory
768 EEG analysis uses OpenNeuro dataset ds005620 (Bajwa et al., 2025); pre-
769 processing and selection are described in Supplemental Material. The repos-
770 itory contains the reproducible analysis and does not redistribute the raw
771 recordings.

772 **Funding and competing interests**

773 This research received no external funding. The author declares no compet-
774 ing interests.

775 Use of AI tools

776 AI assistance to this work was substantial and is not separable by task. Sev-
777 eral coding assistants and chat interfaces were used in an interleaved way
778 throughout: Google’s Antigravity CLI (Gemini 3.1 Pro, Claude 4.6 Opus,
779 Claude 4.6 Sonnet), OpenAI’s Codex CLI (GPT-6 Astra; GPT-5.6 Terra,
780 Luna and Sol), Claude Code (Claude Opus 5), Gemini Pro through its web in-
781 terface, and Anthropic’s Fable and DeepSeek v4 Flash through OpenRouter,
782 accessed with the Hermes CLI. Between them they contributed to the Lean
783 development, the simulation and analysis code, and the drafting and revision
784 of both documents. Because the tools were used in combination and in alter-
785 nation, no section, proof, figure or result is attributable to any one of them,
786 and none is claimed to be. The author is responsible for all content, including
787 everything a tool produced. Machine checking establishes the stated Lean
788 results under their hypotheses; it does not validate biological interpretations
789 or replace scientific review of generated code and prose.

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